Analysis of Algorithms (BDSF25M/A)

Date: *10-Mar-2026* **Quiz 02** **Time:** *15 min*

Roll No:______________**Name:**______________________________

Grading	CLO-1	CLO-2	CLO-3	CLO-4
	/	/	/	/

Read the guidance (similar as in class session) provided on page overleaf and solve following recurrences with only necessary steps. *Note: in (1) 1 is n^0 and in (ii) n is n^1*

(1) $$T(n) = 3T\left(\frac{n}{3}\right) + 1$$

(2) $$T(n) = 5T\left(\frac{n}{2}\right) + n$$

(3) $$T(n) = 3T\left(\frac{n}{2}\right) + n^2$$

(4) $$T(n) = 27T\left(\frac{n}{3}\right) + n^3$$

Solving $T(n) = aT(n/b) + n^k$ via Backward Substitution

1. The Iterative Expansion

We start with the given recurrence and repeatedly substitute the definition of T into itself to observe the developing pattern.

Step 1:

$$T(n) = aT\left(\frac{n}{b}\right) + n^k$$

Step 2: Substitute $T\left(\frac{n}{b}\right) = aT\left(\frac{n}{b^2}\right) + \left(\frac{n}{b}\right)^k$:

$$T(n) = a\left[aT\left(\frac{n}{b^2}\right) + \left(\frac{n}{b}\right)^k\right] + n^k$$

$$T(n) = a^2T\left(\frac{n}{b^2}\right) + a\frac{n^k}{b^k} + n^k$$

Step 3: Substitute $T\left(\frac{n}{b^2}\right) = aT\left(\frac{n}{b^3}\right) + \left(\frac{n}{b^2}\right)^k$:

$$T(n) = a^2\left[aT\left(\frac{n}{b^3}\right) + \left(\frac{n}{b^2}\right)^k\right] + a\frac{n^k}{b^k} + n^k$$

$$T(n) = a^3T\left(\frac{n}{b^3}\right) + a^2\frac{n^k}{b^{2k}} + a\frac{n^k}{b^k} + n^k$$

2. General Form After i Steps

From the steps above, we can generalize the recurrence for i iterations:

$$T(n) = a^iT\left(\frac{n}{b^i}\right) + n^k\sum_{j=0}^{i-1}\left(\frac{a}{b^k}\right)^j$$

3. Reaching the Base Case

The substitution stops when the subproblem size $\frac{n}{b^i} = 1$, which implies:

$$i = \log_b n$$

Assuming the base case $T(1) = c$, we substitute i back into the general form:

$$T(n) = a^{\log_b n} \cdot c + n^k \sum_{j=0}^{\log_b n - 1}\left(\frac{a}{b^k}\right)^j$$

Using the identity $a^{\log_b n} = n^{\log_b a}$:

$$T(n) = c \cdot n^{\log_b a} + n^k \sum_{j=0}^{\log_b n - 1}\left(\frac{a}{b^k}\right)^j$$